Robot Kinematics

Topics

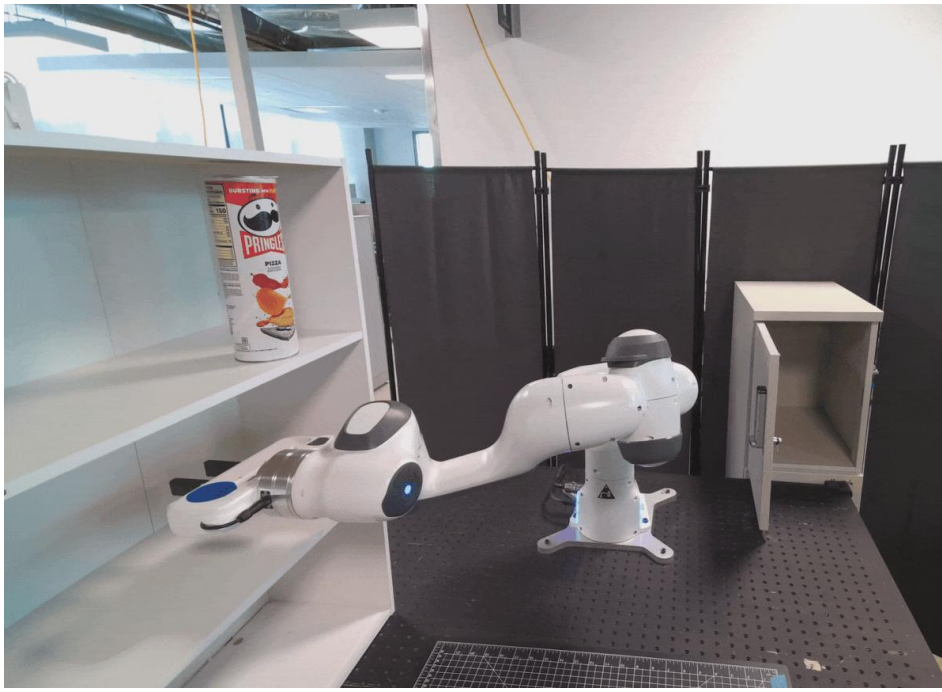
- Forward and Inverse Kinematics of Multi-bodied Robots

Complementary Reading: J.J. Craig, Introduction to Robotics, Chapter 3,4.

Motion Modeling between Mobile Robots and Manipulators



- Geometry and mass distribution remains constant
- Sensing remains fixed relative to base



- Articulated Robot
- Volume changes
- Mass distribution changes
- Sensing (may) change relative to base
- Robot often occludes external sensing

Kinematics

Kinematics: A set of functions describing the geometric position of a robot given joint angles.

- Maps "joint" angles $\boldsymbol{q}$ to position $\boldsymbol{x}$ in Cartesian space.
  - Kinematics can describe positions of *all* the bodies of the robot
- q measures the movement of *one* degree of freedom of the robot (angle, displacement)
- By analytically taking derivatives, it maps speeds and acceleration $\dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$ to speed and acceleration of robot manipulator $\dot{\boldsymbol{x}}, \ddot{\boldsymbol{x}}$. This is *differential kinematics.*
  - The notion of multiple coordinate frames becomes especially useful.
- Measures purely geometric functions and its derivatives,
  - There are no *phenomenological parameters* such as mass, stiffness, force, torque, inertia, damping, friction, etc.
  - Dynamics modeling (i.e., the consideration of force, torque, inertia, friction, etc.) **requires kinematics models** first.

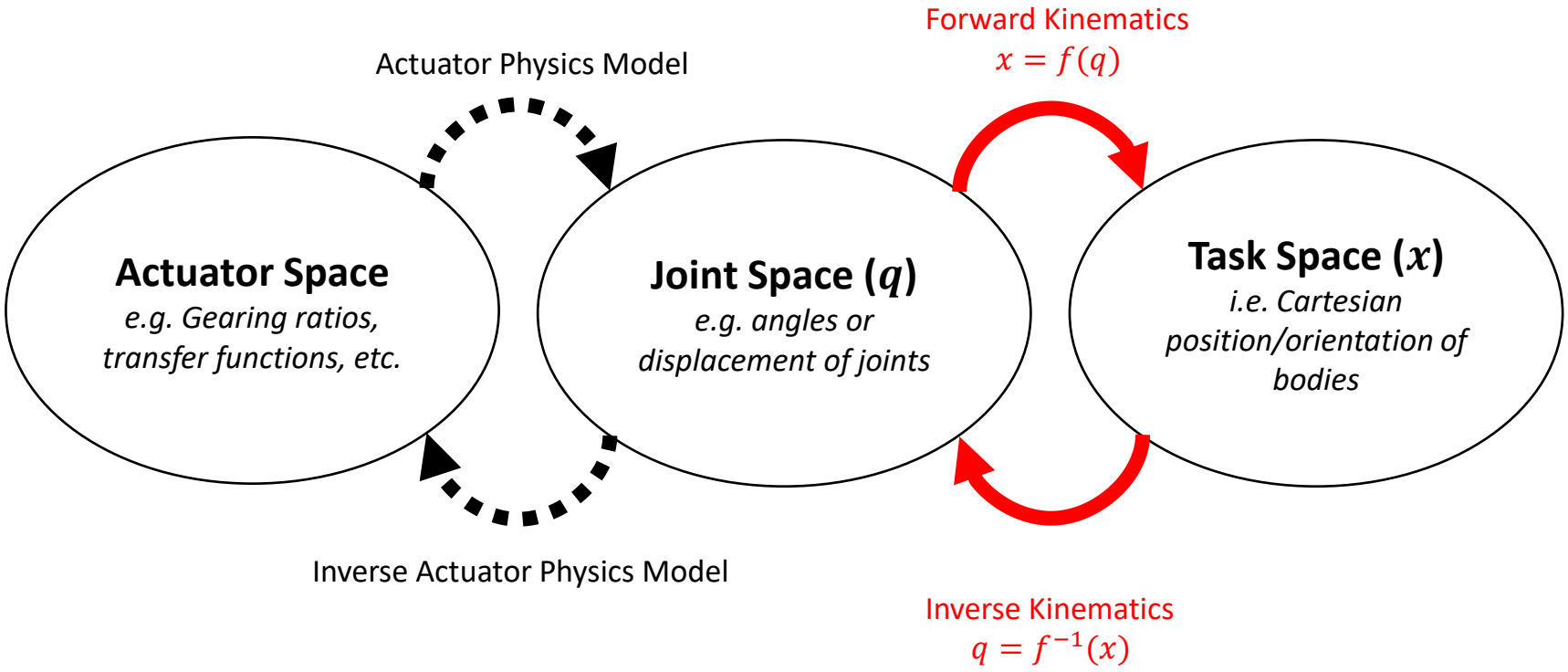
We are interested in two kinematics topics

- Forward Kinematics (angles to Cartesian position)
  - *Given:*
    - The shape of each link and joint locations
    - The angle/displacement of each joint
  - *Find:*
    - The position/orientation of any location on the robot
      - but typically, it's the end effector
- Inverse Kinematics (Cartesian position to angles)
  - *Given:*
    - The shape of each link and joint locations
    - The position/orientation of any location in Cartesian space

    Find:
    - The angles of each joint needed to obtain that position from a given location on the robot (typically the end-effector)

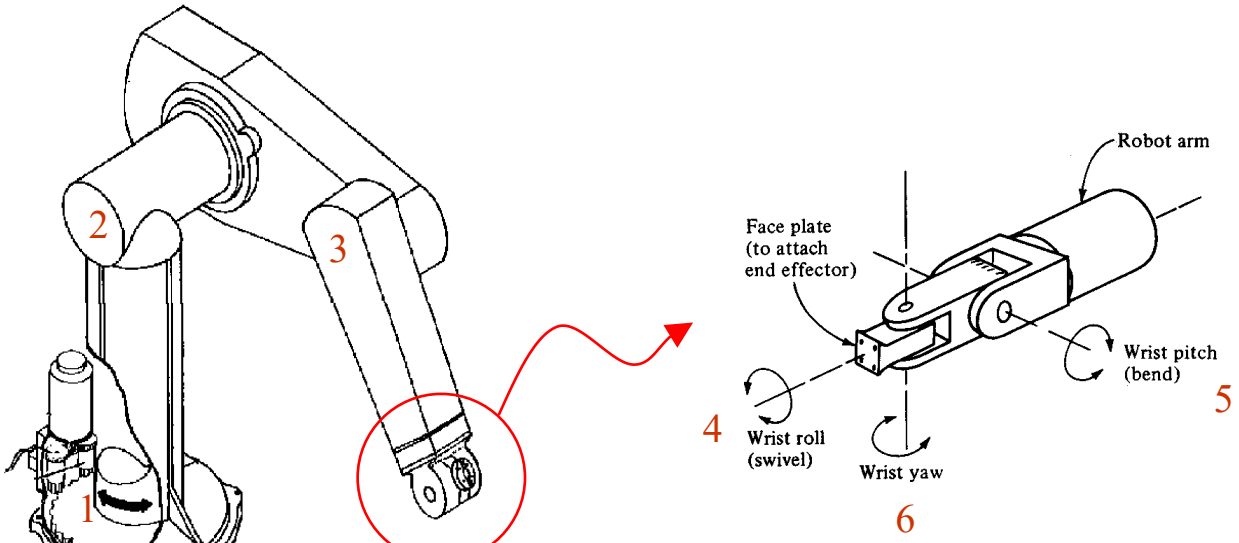


High-level Kinematics Overview



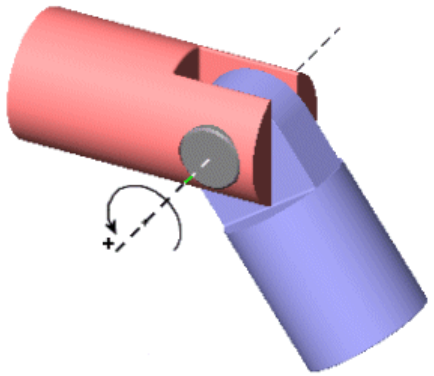
Robot Manipulators

- A joint connects two rigid bodies together
  - i.e, a revolute joint has ONE degree of freedom (1 DOF) that changes the angle between two bodies around a single axis.
  - A prismatic joint has 1 DOF that changes the translation between bodies along a single axis.
- Example: The PUMA 560 has SIX revolute joints

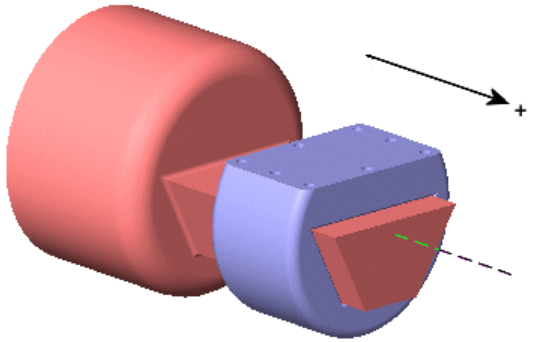


Basic Joints (R,P,S)

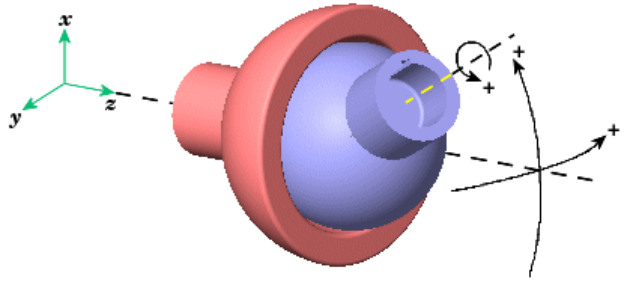
- **R**evolute Joint: 1 DOF



- **P**rismatic Joint: 1 DOF (linear)

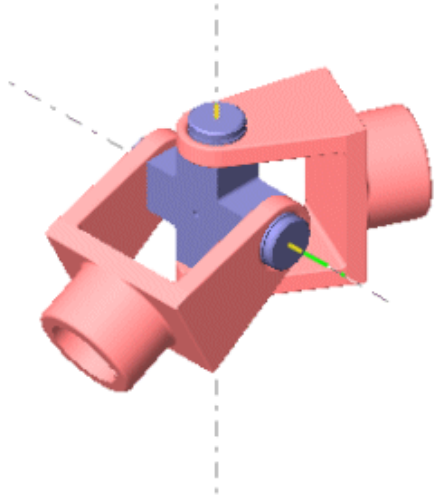


- **S**pherical Joint: 3 DOF



Most other joints are compositions of the previous joints

- Two Revolute Joints (θ_1, θ_2)



- Example (6-DOF arm)



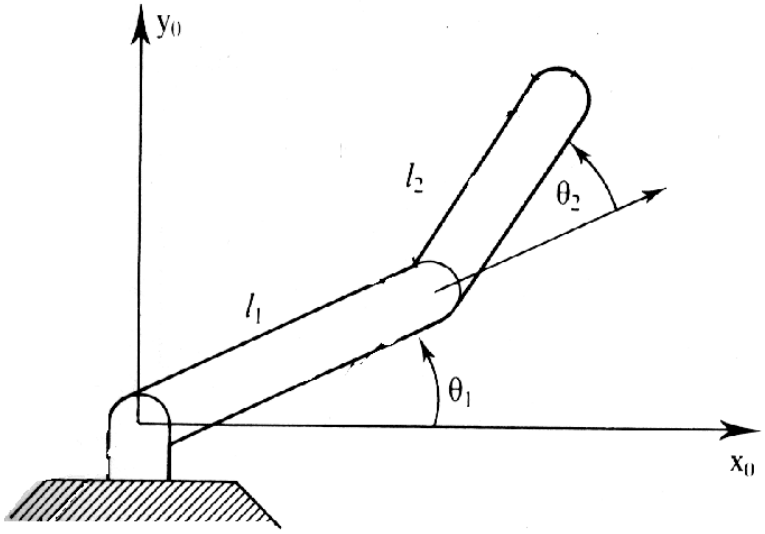
2D robot manipulator example

Forward kinematics problem:

You have a robotic arm that starts out aligned with the x_o-axis.

- Set link 1 to move by q_1
- Set link 2 to move by q_2

What is the position of the end of the robotic arm?



- **Geometric Solution**
  - no coordinates, no vectors, pure geometry
  - Easy for very low dimensional spaces
  - Non-trivial for higher-dimensional spaces
  - Non-generalizable

- **Algebraic Solution**
  - vector algebra approach
  - Involves coordinate systems and vector representations
  - Generalizable to any multi-body dynamical system

Forward Kinematics

1. Choose a "World" Frame $\{C_0\}$
2. Place a coordinate frame on every Link, $\{C_1\} \ldots \{C_N\}$ where $\{R_N, p_N\}$ is the orientation and position of the end-effector.
3. Describe all frames $\{C_i\}$ relative to the previous frame $\{C_{i-1}\}$ in the series using a 4x4 homogenous matrix, e.g., ${}^{i-1}_{i}H := \begin{bmatrix} {}^{i-1}_{i}R & {}^{i-1}p_i \\ 0 & 1 \end{bmatrix}$
3. Starting with the Link N, describe it in the previous frame $\{C_{N-1}\}$ with a 4x4 matrix:

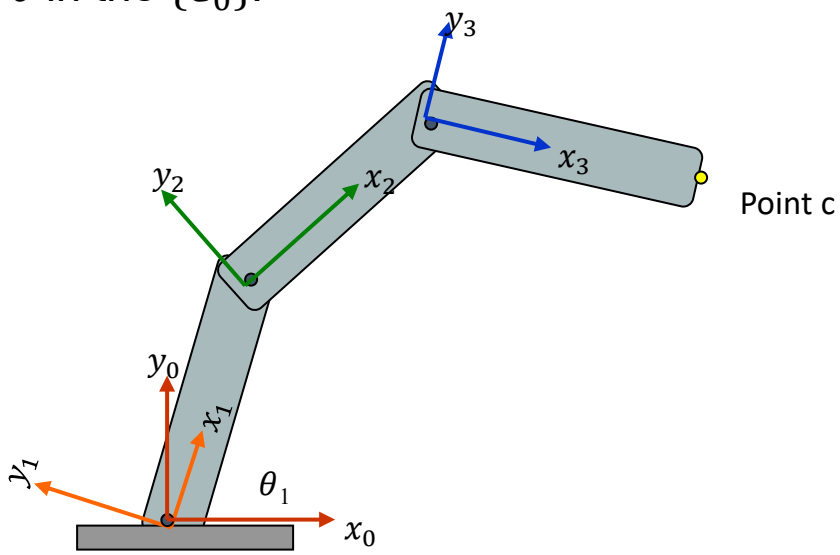
$$ {}^{0}\{C_1\} = \begin{bmatrix} {}^{N-1}_{N}R & {}^{N-1}p_N \\ 0 & 1 \end{bmatrix} = {}^{N-1}_{N}H $$

4. Work your way backward to $\{C_0\}$ with successive homogenous transforms

$$ {}^{0}_{N}H = {}^{0}_{1}H \, {}^{1}_{2}H \ldots {}^{N-2}_{N-1}H \, {}^{N-1}_{N}H = \begin{bmatrix} {}^{0}_{N}R & {}^{0}p_N \\ 0 & 1 \end{bmatrix} $$

Example Problem:

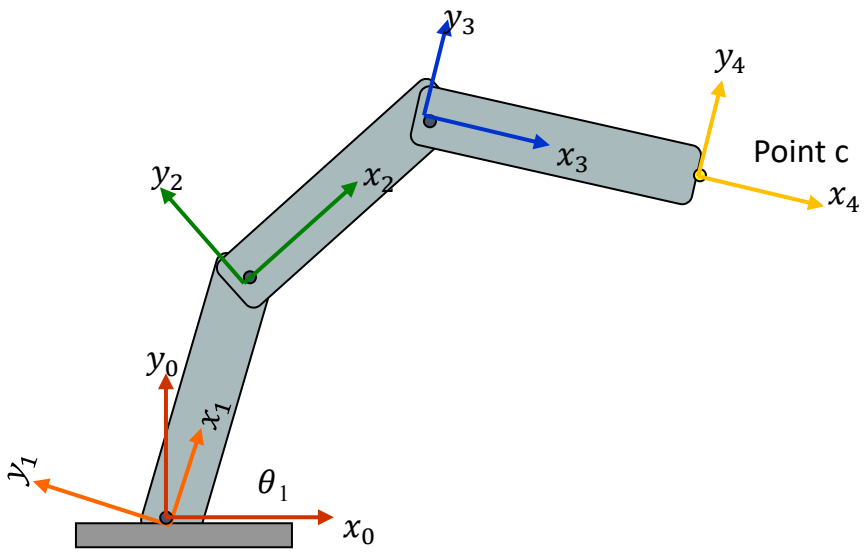
Consider a three-link arm with link lengths l_1, l_2, l_3, respectively. Describe position of the yellow dot at point c in the $\{C_0\}$.



- Choose coordinate frames (z-axes on joints, x pointing towards next joint)
- The position of the point c relative to the $\{C_3\}$ frame is ${}^3p_c = [l_3, 0]^\top$
- Following the homogenous transforms from $\{C_3\}$ to $\{C_0\}$

$$
\begin{aligned}
{}^0p_c &= {}^0_1H\ {}^1_2H\ {}^2_3H[{}^3p_c \quad 1]^\top \\
&= \begin{bmatrix} {}^0_1R & {}^0p_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^1_2R & {}^1p_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^2_3R & {}^2p_3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^3p_c \\ 1 \end{bmatrix}
\end{aligned}
$$

- **Slight variation on the last solution**:
  - Make the yellow dot the origin of a new coordinate $\{C_4\}$ frame.
  - Orientation can be arbitrarily chosen (x typically pointing outwards)



$$ {}^0_4H = {}^0_1H\ {}^1_2H\ {}^2_3H\ {}^3_4H $$

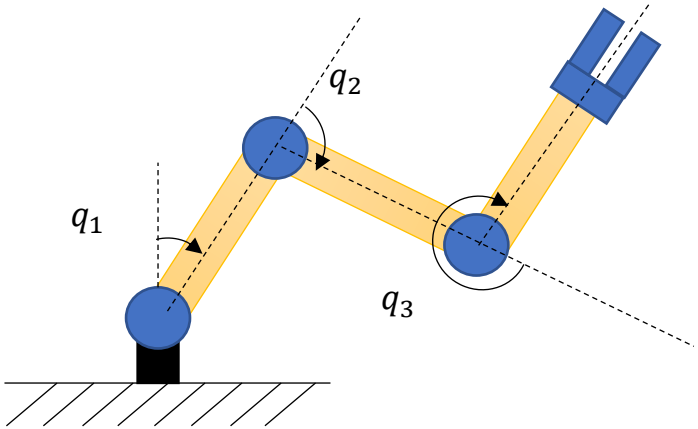
Where is Point C, i.e. where is ${}^0p_c$?

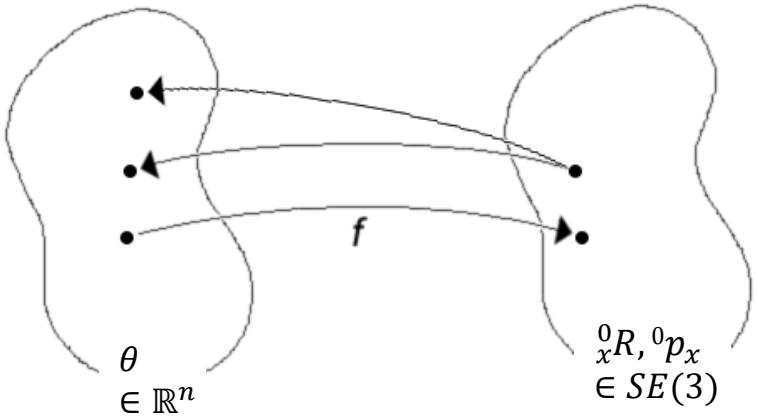
Inverse Kinematics

- Consider the forward kinematics of a robot, that maps $q \rightarrow {}^{0}_{x}R, {}^{0}p_{x}$
- Inverse kinematics maps:

$$ {}^{0}_{x}R, {}^{0}p_{x} \rightarrow q $$

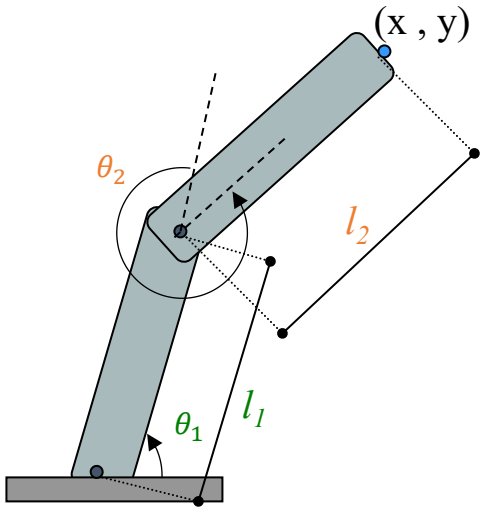
- Inverse kinematics can result in infinite, several, a single, or no solution.
- No standard analytical approach to solving inverse kinematics
  - Iterative approach used, and no guarantee of convergence





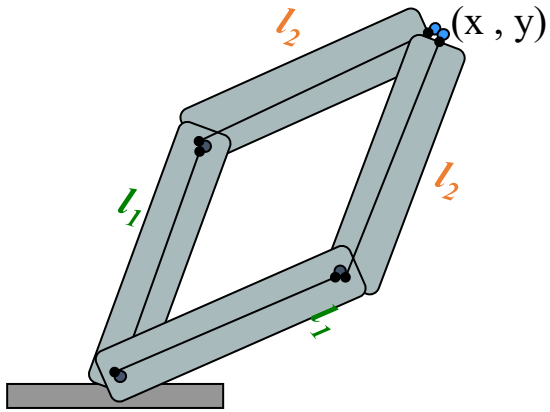
Example: Two link manipulator, and l_1, l_2, x, y,

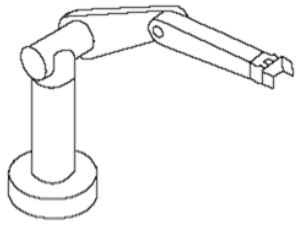
Find θ_1, θ_2

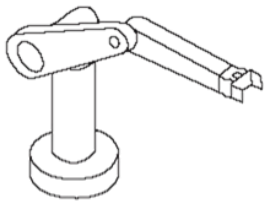


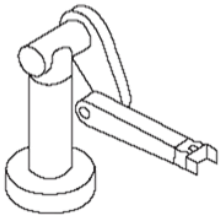
Inverse problem is non-exact:

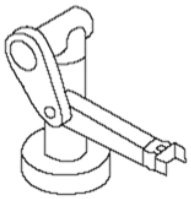
- Multiple solutions exist
- Often called **arm up** and **arm down** configurations.
- Number of solutions expand exponentially with degrees of freedom.




Left Arm Elbow Up

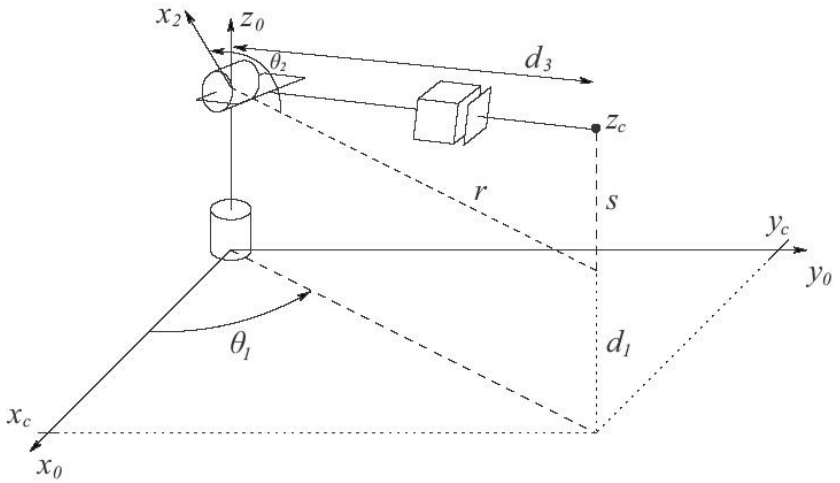

Right Arm Elbow Up


Left Arm Elbow Down


Right Arm Elbow Down

Example: RRP manipulator

- "RRP" describes rotary-rotary-prismatic sequence of joints



Analytical Inverse Kinematics: given x_c, y_c, z_c find θ_1, θ_2, d_3

- Solve for θ_1:

$$\theta_1 = arctan2(y_c, x_c)$$

- Solve for θ_2:

$$\theta_2 = arctan2(s, r)$$

$$r = \sqrt{x_c^2 + y_c^2}$$

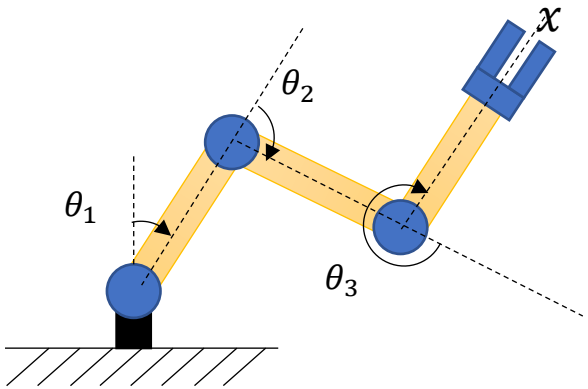
$$s = z_c - d_1$$

- Solve for d_3:

$$d_3 = \sqrt{r^2 + s^2} = \sqrt{x_c^2 + y_c^2 + (z_c - d_1)^2}$$

Example: planar RRR manipulator

How many solutions to reach point x?



- given $x \in \mathbb{R}^2$, the forward kinematics can be written as:

$$x_1 = a_1 c\theta_1 + a_2 c\theta_{1+2} + a_3 c\theta_{1+2+3}$$
$$x_2 = a_1 s\theta_1 + a_2 s\theta_{1+2} + a_3 s\theta_{1+2+3}$$

… the problem is under-constrained (two equations and three unknowns)

A general solution to inverse kinematics

Construct a minimization problem, for example:

$$\underset{\theta}{\text{minimize}} \quad \left\| {}^{0}_{N}H(\theta)\, {}^{N}p_x - {}^{0}p_x \right\|$$

${}^{0}p_x$: desired position of object x in hand

${}^{N}p_x$: position of object relative to the pose of the end effector coord. frame $\{C_N\}$

${}^{0}_{N}H(\theta){}^{N}p_x$: position of object x based on forward kinematics.

- Solve using **gradient descent.**

Algorithm Method of Gauss-Newton

input : $f : \mathbb{R}^n \to \mathbb{R}$ a function such that $f(\mathbf{x}) = \sum_{i=1}^{m} (f_i(\mathbf{x}))^2$
where all the f_i are differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$
$\mathbf{x}^{(0)}$ an initial solution

output: $\mathbf{x}^\star$, a local minimum of the cost function f.

begin
  $k \leftarrow 0$;
  while STOP-CRIT **and** $(k < k_{max})$ **do**
    $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \boldsymbol{\delta}^{(k)}$;
    with $\boldsymbol{\delta}^{(k)} = \arg\min_{\boldsymbol{\delta}} \|\mathcal{F}(\mathbf{x}^{(k)}) + \mathbf{J}_{\mathcal{F}}(\mathbf{x}^{(k)})\boldsymbol{\delta}\|^2$;
    $k \leftarrow k + 1$;
  return $\mathbf{x}^{(k)}$
end

Q: what is the “gradient” in gradient descent doing?

Levenberg Marquardt (LM)

- A robust gradient descent method, more useful than G-N.
- Has dynamic adjustment of step size, using parameter λ to converge faster, reduce overshoot.

Algorithm 5: Levenberg-Marquardt algorithm

input : $f : \mathbb{R}^n \to \mathbb{R}$ a function such that $f(\mathbf{x}) = \sum_{i=1}^{m} (f_i(\mathbf{x}))^2$
where all the f_i are differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$
$\mathbf{x}^{(0)}$ an initial solution

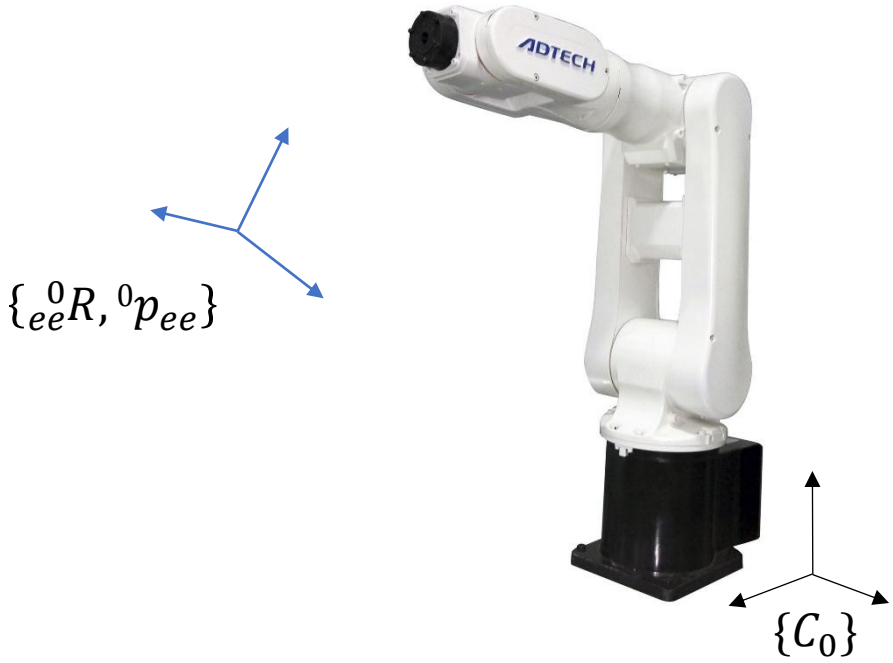
output: $\mathbf{x}^\star$, a local minimum of the cost function f.

```
1  begin
2      k ← 0 ;
3      λ ← max diag(J^T J) ;
4      x ← x^(0) ;
5      while STOP-CRIT and (k < k_max) do
6          Find δ such that (J^T J + λ diag(J^T J))δ = J^T f ;
7          x' ← x + δ ;
8          if f(x') < f(x) then
9              x ← x' ;
10             λ ← λ/ν ;
11         else
12             λ ← νλ ;
13         k ← k + 1 ;
14     return x
15 end
```

Q: what is lambda calculating?

Nuanced challenges with inverse kinematics:

What if the desired position includes orientation? i.e. we want to match the end-effector coordinate frame $\{C_{ee}\}$ to a desired orientation ${}^{0}_{ee}R = {}^{0}\{u_x, u_y, u_z\}_{ee}$ and position ${}^{0}p_{ee}$. **Define the residual function.**



Q: Consider if the solution cannot be minimized to 0. What will happen?

Other nuances with Inverse Kinematics

- Not all robots have analytical inverse kinematics
- robot has modeling errors (ϵ) or deficiencies
- When there are constraints on the inputs

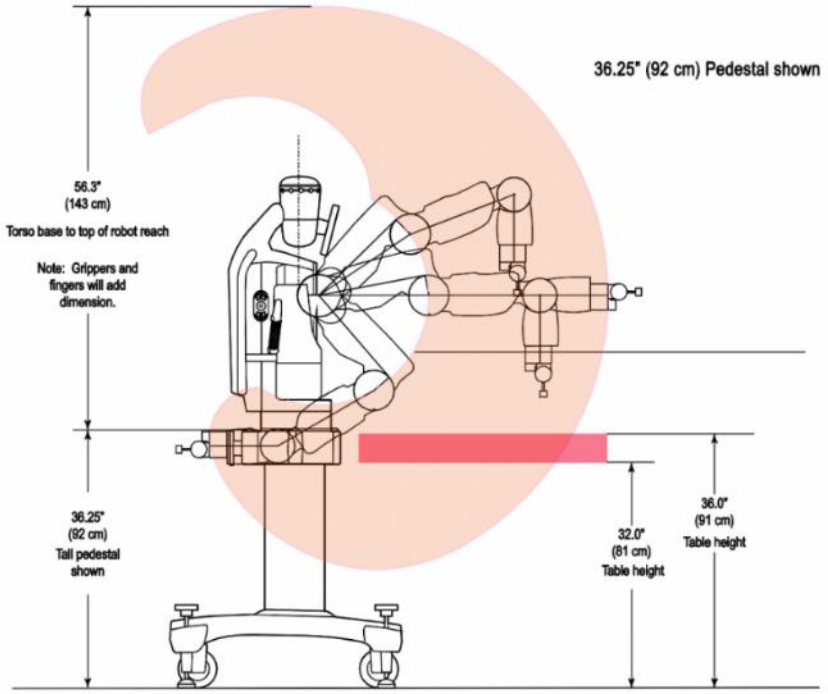
$$c_{neq}(q) \leq 0$$

$$c_{eq}(q) = 0$$

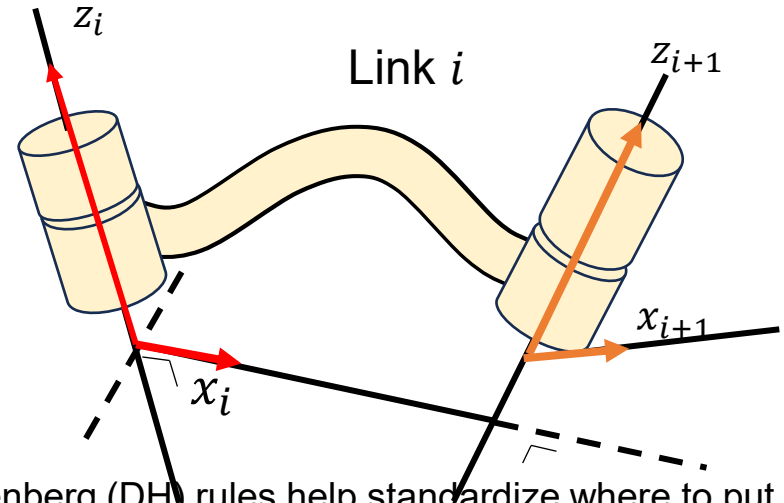
- When your desired workspace exceeds your reachable workspace

$$x \in X_{reachable}$$

- When the environment influences feasibility of movement (i.e. collisions)



A Standard for Drawing Coordinate Frames: (modified) Denavit-Hartenberg Method



The Modified Denavit-Hartenberg (DH) rules help standardize where to put coordinate frames. Follow below:

i. **Draw axis of motion on all joints**. Let this axis be the z axis.
   - *Assign direction of axis based on what is +ve rotation for the robot joints, if known.*

ii. **Find a common normal (shortest line) between axes**.
   - *If axes are parallel, the common normal can slide anywhere on the axes. Place it at a location that "makes sense" to you for now.*

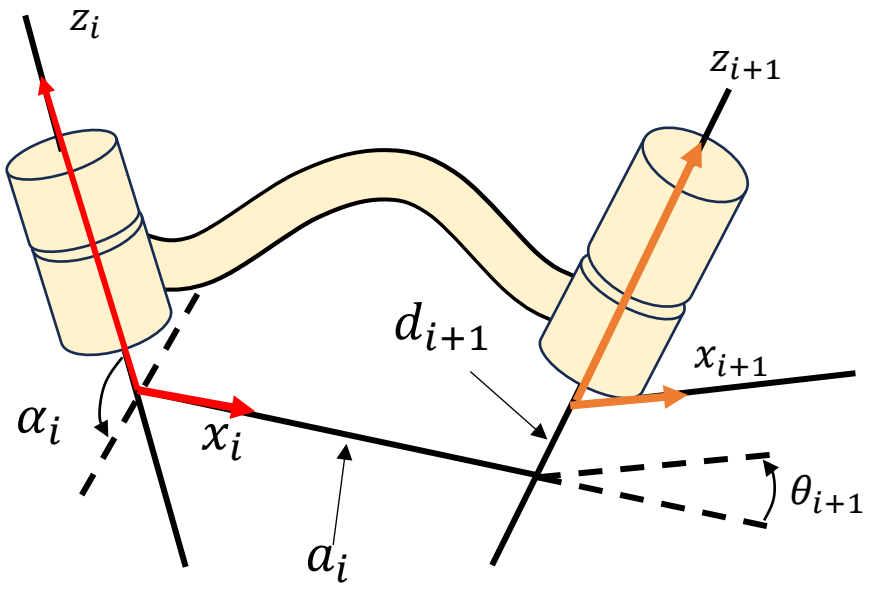
iii. **For successive frames (from tip to base), draw x_i such that it points towards the z_{i+1}**, along its common normal.

iv. **Origin of the frame is where x intersects the next z axis.**
   - If you had any parallel axes from (ii) that you had previous assigned that has put the origin in an awkward place, now would be the time to move the common normal to a better location so that you minimize unnecessary offsets.

D-H parameters explained

Four parameters for each transformation

- a **(link length):** Distance along the X-axis between two consecutive Z-axes.
- α **(link twist):** Angle between the two consecutive Z-axes, measured around the X-axis.
- d **(link offset):** Distance along the Z-axis between the origins of two consecutive coordinate frames.
- θ **(joint angle):** Rotation about the Z-axis to align the X-axis of one frame with the next.



The D-H parameters describe the successive homogenous matrices as:

$$
{}^{i-1}_{i}H = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & a_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\theta_i)\cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & -d_i \sin(\alpha_{i-1}) \\ \sin(\theta_i)\sin(\alpha_{i-1}) & \cos(\theta_i)\sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & d_i \cos(\alpha_{i-1}) \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Try to do this yourself:



Non Serial-Chain Robot Manipulators:



Continuum Robots:

